

Demo: Kelvin Water Dropper

Electrostatic feedback without a spark

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

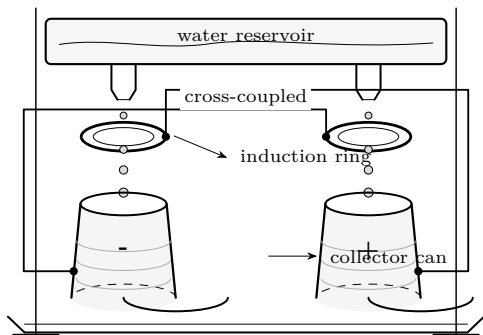
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- Adult-built paired drip apparatus
- Two metal cans and two induction rings
- Plastic tubing
- Water tray
- Electroscope
- Grounding lead used only after flow stops

Predict first

How can initially neutral water and cans separate charge without a battery?



A tiny imbalance reinforces itself: each can biases the opposite falling stream.

Investigate

1. Adult checks stable supports, dry floor, spill tray, and generous ring separation. Operate indoors only, never during thunderstorm conditions.
2. Start two smooth drips and watch the electroscope connected by the apparatus design.
3. Stop flow well before any spark, snapping, hair raising, or painful sensation. Do not approach conductors while operating.
4. With water stopped, adult discharges through the intended high-resistance path and verifies zero indication before touch.

Explain from the evidence

A tiny initial imbalance polarizes droplets; cross-coupling reinforces charge separation through positive feedback. Gravitational energy of falling water ultimately supplies the electrical energy. This edition uses the device only as an adult-built, no-spark electrostatic demonstration. It is not a learner wiring project.

Change one thing

Vary drip rate while keeping geometry fixed and compare time to a conservative electroscope mark; never use spark length as a measurement.

Evidence record	
Prediction	_____
One change	_____
Observation	_____
Claim + because	_____

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.

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